

UX Design Detailed Study Notes

Topic 1 — Information Architecture (IA)

What is Information Architecture?

Information Architecture (IA) is the process of organising, structuring, and labelling content in a product so users can easily navigate and find information.

It acts like the “blueprint” or “skeleton” of a website/app.

Good IA improves:

- Findability
- Navigation
- User understanding
- Efficiency
- User satisfaction

Bad IA causes:

- Confusion
 - Frustration
 - Users getting lost
 - Increased cognitive load
-

3 Key IA Models (NNgroup)

1. Navigation Model

The visible structure users interact with.

Examples:

- Menus
- Navigation bars
- Sidebars
- Tabs

- Links

Purpose:

Helps users move around the product.

Example:

Amazon categories menu.

2. Taxonomy Model

How content is grouped and categorised.

Examples:

- Product categories
- Blog tags
- File folders

Purpose:

Makes information easier to understand and locate.

Example:

Clothing website categories:

- Men
 - Women
 - Kids
-

3. Full IA Structure

The complete map of all screens/pages and their relationships.

Includes:

- Parent pages
- Child pages
- Connections between screens

Purpose:

Helps designers understand the whole system structure.

Core IA Principles

1. Organisation

Content should be logically arranged.

Example:

Grouping settings into:

- Privacy
 - Notifications
 - Account
-

2. Labelling

Clear and familiar names for menus/buttons.

Good:

“Contact Us”

Bad:

“Connect Experience Portal”

3. Navigation

Users should always know:

- Where they are
 - Where they can go
 - How to return
-

4. Search

Important for content-heavy platforms.

Good search:

- Fast
 - Accurate
 - Filters available
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Common IA Patterns

1. Hierarchical Structure

Tree-like structure.

Example:

Home → Products → Electronics → Phones

Most common pattern.

2. Sequential Structure

Step-by-step flow.

Example:

Checkout process.

3. Matrix Structure

Multiple navigation pathways.

Example:

Netflix browsing by:

- Genre
 - Trending
 - Search
 - Recommendations
-

4. Organic Structure

Free exploration with no fixed path.

Example:

Art portfolio websites.

Why IA Matters

Good IA:

- Reduces confusion
 - Improves usability
 - Helps users complete tasks faster
 - Supports accessibility
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Topic 2 — Task Flow and User Flow

Task Flow

Definition

A Task Flow shows the exact steps needed to complete ONE task.

Features:

- Linear
- Simple
- No branching
- No personas

Example:

Steps to order food:

1. Open app
2. Select restaurant

3. Add items
 4. Checkout
 5. Payment
-

User Flow

Definition

A User Flow maps how a specific user type moves through a product.

Features:

- Includes decisions
- Multiple paths
- Entry points
- Exit points
- User goals

Example:

If payment fails:

- Retry payment
 - Change method
 - Cancel order
-

Difference Between Task Flow and User Flow

Task Flow	User Flow
Simple	Complex
One path	Multiple paths
No decisions	Includes decisions
Focus on task	Focus on user journey

Why Flows Matter

Flows help designers:

- Identify friction
 - Detect unnecessary steps
 - Improve usability
 - Plan interfaces before UI design
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Topic 3 — Cognitive Load

Definition

Cognitive load is the amount of mental effort required to use a system.

Goal of UX:

Reduce unnecessary mental effort.

Types of Cognitive Load

1. Intrinsic Load

Natural difficulty of the task.

Cannot fully remove it.

Example:

Learning coding.

UX goal:

Simplify and guide users.

2. Extraneous Load

Mental effort caused by poor design.

Examples:

- Cluttered screens
- Confusing labels
- Too many options
- Inconsistent navigation

This SHOULD be removed.

3. Germane Load

Mental effort used for learning.

Helpful cognitive effort.

Example:

Learning how a dashboard works.

Good UX supports this.

Techniques to Reduce Cognitive Load

Chunking

Break information into smaller groups.

Example:

Phone number formatting.

Progressive Disclosure

Show advanced info only when needed.

Example:

“Advanced settings” dropdown.

Familiar Patterns

Use common UI conventions.

Example:

Shopping cart icon.

Minimise Choices

Too many choices overwhelm users.

Related concept:

Miller’s Law (7 ± 2 items).

Topic 4 — Storyboarding and Wireframes

Storyboards

Definition

Visual stories showing the user experience.

Helps understand:

- Context
- Emotions
- Problems
- Goals

Types of Storyboards

1. Big Picture Storyboard

Focus:

- User emotions
- Environment
- Motivation

Shows WHY users need the product.

2. Close-Up Storyboard

Focus:

- Screens
- Interactions
- Product usage

Shows WHAT users do.

4 Elements of Storyboards

Character

The user/persona.

Scene

The environment/context.

Plot

The sequence of actions.

Narrative

Overall story/message.

Wireframes

Definition

Simple layouts showing structure and functionality.

Wireframes focus on:

- Placement
- Navigation
- Layout
- User flow

NOT:

- Colours
 - Final visuals
 - Branding
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Types of Wireframes

Paper Wireframes

Quick hand-drawn sketches.

Cheap and fast.

Digital Wireframes

Created using tools like:

- Figma
 - Adobe XD
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UX Process Flow

Research → Personas → User Stories → Storyboards → Wireframes → Prototypes

Topic 5 — Low-Fidelity Prototypes

Definition

A low-fidelity prototype is a simple interactive version of a product with minimal visual detail.

Purpose:

Early testing and feedback.

Types

1. Paper Prototype

Physical sketches.

Interactions simulated manually.

2. Digital Lo-Fi Prototype

Clickable wireframes using software.

Example:

Figma prototype links.

Why Use Lo-Fi Prototypes?

Fast

Quick to create.

Cheap

Low cost.

Easy to Change

No emotional attachment to designs.

Focus on Usability

Users focus on functionality instead of visuals.

Lo-Fi vs Hi-Fi

Lo-Fi	Hi-Fi
Basic visuals	Real visuals
Wireframes	Final UI
Early testing	Final testing
Fast edits	More expensive

Topic 6 — Mental Models

Definition

Mental models are users' expectations about how something should work based on past experiences.

Why Important?

Users compare your product with:

- Other apps
- Physical objects
- Previous habits

Good UX matches expectations.

Examples

Desktop Metaphor

Computer desktops mimic physical desks.

Shopping Cart Icon

Users expect it to store selected products.

Jakob's Law

Users spend most of their time on other websites.

Therefore:

They expect your website to work similarly.

Mismatched Mental Models

When products behave unexpectedly:

- Confusion increases
 - Learning becomes harder
 - Users feel frustrated
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Good UX Strategy

Either:

- Match existing mental models
OR
 - Introduce new concepts gradually with onboarding
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Topic 7 — Inclusive and Accessible Design

Accessible Design

Definition

Designing products usable by people with disabilities.

Focus:

Removing barriers.

Uses:

WCAG guidelines.

Examples:

- Screen reader support
 - Keyboard navigation
 - Alt text
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Inclusive Design

Definition

Designing for diverse human experiences from the start.

Considers:

- Age
 - Language
 - Culture
 - Gender
 - Economic status
 - Ability
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Key Difference

Accessibility	Inclusive Design
Reactive	Proactive
Focus on disabilities	Focus on all diversity
Compliance-focus	Human-focused

Curb Cut Effect

Accessibility features often help everyone.

Example:

Wheelchair ramps also help:

- Parents with strollers
 - Cyclists
 - Delivery workers
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Why Inclusive Design Matters

Benefits:

- Larger audience
 - Better usability
 - Ethical design
 - Better business outcomes
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Topic 8 — Visual Design Principles

1. Emphasis

Makes important elements stand out.

Methods:

- Size
- Colour
- Contrast
- Placement

Example:

CTA buttons.

2. Hierarchy

Shows importance order visually.

Methods:

- Font size
- Weight
- Position

Example:

Large headlines.

3. Scale and Proportion

Relative sizing of elements.

Helps:

- Balance layouts
 - Show importance
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4. Unity

Everything feels connected and consistent.

Achieved through:

- Consistent colours
 - Typography
 - Spacing
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5. Variety

Adds visual interest.

Prevents boring layouts.

Gestalt Principles

Proximity

Close items seem related.

Similarity

Similar items seem connected.

Continuity

Eyes naturally follow lines and paths.

Closure

People complete incomplete shapes mentally.

Figure/Ground

Distinguishing foreground from background.

Topic 9 — Error State Design

Definition

Error states inform users when something goes wrong and help them recover.

Principles of Good Error Design

1. Be Specific

Good:

“Password must be at least 8 characters.”

Bad:

“Error occurred.”

2. Use Human Language

Avoid technical jargon.

Bad:

“HTTP 403.”

Good:

“You don’t have permission to access this page.”

3. Offer a Solution

Tell users what to do next.

Example:

“Check your internet connection and retry.”

4. Timing Matters

Inline validation should appear after users finish typing.

5. Don’t Rely Only on Colour

Use:

- Icons
- Text
- Labels

Accessibility matters.

Types of Error States

Form Validation Errors

Incorrect input.

Empty States

No data available.

Example:

“No messages yet.”

Network Errors

Internet/connectivity issues.

Permission Errors

User lacks access rights.

System Errors

Backend/system crashes.

Nielsen's Error Heuristics

Good errors should:

- Help users recognise the problem
- Explain clearly
- Suggest recovery steps
- Use plain language